

Merrivoss Bioanalysis Inc

KT12 Permeability, Solubility and Plasma Binding

Final study report R1

Study MV-2008-013

Accession M08/BIO/00303

Prepared for Kelbravik Therapeutics ApS

Dr M. Larkin, DMPK Study Director

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1 Scope and study disposition

The KT12 test article set comprised 32 compounds. Work opened on 02/18/2008 and the interim communication was issued on 03/05/2008. This report closes the study on 03/21/2008 for Kelbravik Therapeutics ApS.

Table 1. Study scope and controlling identifiers

Metric	Value
Project	MV-2008-013
Client	Kelbravik Therapeutics ApS
Compounds tested	32
Series / chemotype	Kelbravik cyclopropyl carboxamide series 12 (Cyclopropyl carboxamide)
Assays	Plasma protein binding (equilibrium dialysis), Caco-2 permeability, Kinetic solubility
Turnaround (calendar days)	32
Price	USD 14,900

Binding, transport and solubility describe separate exposure constraints. The matrices and analytical limits are stated before the branch decision. No oral exposure measurement is supplied by this decision set.

2 fu,p by equilibrium dialysis

Human plasma was assessed by equilibrium dialysis for 4 h at 37 °C. Bound percentage, unbound percentage and recovery are retained together. The unbound column must be converted to a fraction before use in a calculation that requires fractional fu,p.

Table 2. Human plasma binding and recovery

Compound	% bound	Unbound (%)	Recovery (%)
KT12-06304	89.9	10.13	92
KT12-06306	53.5	46.55	94
KT12-06310	93.0	7.01	104
KT12-06312	62.1	37.91	98
KT12-06316	45.9	54.06	83
KT12-06336	49.9	50.15	95
KT12-06340	51.6	48.35	100
KT12-06346	63.2	36.82	97
KT12-06348	62.6	37.43	85
KT12-06354	52.2	47.79	87
KT12-06366	85.7	14.30	92
KT12-06372	87.4	12.61	94
KT12-06384	60.0	40.04	102
KT12-06396	89.7	10.27	89
KT12-06402	83.7	16.33	90
KT12-06408	57.9	42.13	104
KT12-06414	58.2	41.76	94
KT12-06424	87.9	12.05	101
KT12-06426	86.5	13.48	97
KT12-06430	65.4	34.64	91
KT12-06436	60.2	39.77	89
KT12-06438	63.8	36.21	98
KT12-06442	69.6	30.36	91
KT12-06450	70.5	29.51	97
KT12-06456	75.6	24.42	100
KT12-06460	72.9	27.08	98
KT12-06462	82.7	17.26	80
KT12-06468	60.4	39.60	104
KT12-06474	65.0	35.01	91
KT12-06480	53.7	46.32	87
KT12-06484	71.7	28.32	102
KT12-06486	54.0	46.04	93

Warfarin control 98-99% bound

Recovery is a material-balance context, not an exposure result. A small unbound fraction must be interpreted with the analytical record rather than taken as evidence of absent free compound.

3 Caco-2 flux and efflux ratio

Bidirectional permeability was assessed at 10 μ M and pH 7.4. Apparent permeability is expressed in 10^{-6} cm/s, while efflux ratio is dimensionless. Both directions and recovery are retained in the table.

Table 3. Bidirectional Caco-2 permeability

Compound	Papp A-B (10^{-6} cm/s)	Papp B-A (10^{-6} cm/s)	Efflux ratio	Recovery (%)
KT12-06304	8.3	51	6.1	94
KT12-06306	24	90	3.7	84
KT12-06310	14	84	5.9	89
KT12-06312	15	31	2.0	100
KT12-06316	32	78	2.4	70
KT12-06336	25	71	2.8	88
KT12-06340	11	48	4.2	98
KT12-06346	41	80	2.0	99
KT12-06348	17	27	1.6	96
KT12-06354	14	52	3.6	93
KT12-06366	19	35	1.9	79
KT12-06372	29	50	1.7	100
KT12-06384	30	55	1.9	70
KT12-06396	7.6	19	2.5	102
KT12-06402	26	46	1.7	93
KT12-06408	15	67	4.5	73
KT12-06414	23	60	2.6	81
KT12-06424	34	48	1.4	92
KT12-06426	31	129	4.2	82
KT12-06430	8.0	42	5.2	74
KT12-06436	25	55	2.2	73
KT12-06438	23	65	2.8	80
KT12-06442	26	47	1.8	91
KT12-06450	9.0	22	2.4	96
KT12-06456	42	133	3.2	99
KT12-06460	24	62	2.6	90
KT12-06462	40	38	0.97	99
KT12-06468	20	39	1.9	99
KT12-06474	24	42	1.7	78
KT12-06480	13	36	2.7	96
KT12-06484	25	57	2.2	94
KT12-06486	28	81	2.9	100

A ratio alone cannot distinguish weak forward entry from increased reverse transport. The complete directional record is needed before assigning a permeability-limited branch.

4 DMSO-dilution solubility

Kinetic solubility was assessed in PBS at pH 7.4 and is expressed in μM . Upper-limit entries remain limit statements rather than exact concentrations.

Table 4. Kinetic solubility in PBS

Compound	Solubility (μM)
KT12-06304	>200
KT12-06306	>200
KT12-06310	16
KT12-06312	>200
KT12-06316	>200
KT12-06336	>200
KT12-06340	>200
KT12-06346	>200
KT12-06348	>200
KT12-06354	>200
KT12-06366	118
KT12-06372	>200
KT12-06384	>200
KT12-06396	>200
KT12-06402	>200
KT12-06408	>200
KT12-06414	>200
KT12-06424	>200
KT12-06426	>200
KT12-06430	>200
KT12-06436	>200
KT12-06438	>200
KT12-06442	>200
KT12-06450	>200
KT12-06456	>200
KT12-06460	134
KT12-06462	>200
KT12-06468	>200
KT12-06474	>200
KT12-06480	>200
KT12-06484	>200
KT12-06486	>200

From 10 mM DMSO stock, 2% DMSO final, nephelometry

Dilution into buffer is an availability challenge. The result is not equilibrium solubility of a selected solid form, a vehicle-performance result or an intestinal dissolution measurement.

5 Purpose at the exposure gate

Kelbravik needed to distinguish limited dissolved availability from limitation at a cellular barrier. The cyclopropyl carboxamide set was therefore evaluated across transport and solubility, with plasma binding retained as a separate exposure context.

A formulation assessment addresses a different question from a permeability follow-up. Keeping the branches distinct prevents every weak forward-flux result from being assigned the same remedy. A binding advantage cannot remove either constraint by itself.

6 Accession-lot identity

The controlling accession is M08/BIO/00303. The companion assay worksheet contains the accession-lot mapping. Complete KT12 compound identifiers join the tables; row position and shortened suffixes are not identity links.

The report assumes consistent submitted identity across the systems. It does not replace material identity or purity characterization. Later formulation work should retain the accession link while identifying any new material lot separately.

7 Matrix and species definitions

Table 5. Matrix boundaries

Assessment	Matrix	Biological basis	Unit
fu,p by equilibrium dialysis	Plasma and buffer	Human plasma	Bound and unbound %
Caco-2 flux and efflux ratio	Cell monolayer compartments	Human-derived model	10 ⁻⁶ cm/s; ratio
DMSO-dilution solubility	PBS pH 7.4	No species matrix	µM

The cell model is not a measured human absorption study. No rat binding result is supplied. Solubility in buffer has no species assignment. These distinctions apply to any later extract of the tables.

8 Dialysis interpretation

The plasma and buffer compartments provide an operational estimate of unbound fraction under the assay conditions. Nonspecific loss or incomplete equilibration can distort that estimate, which is why recovery remains alongside binding.

Highly bound material is not automatically unsuitable for progression. Its relevance depends on the eventual exposure and concentration context. Neither is measured by the dialysis experiment, and no free plasma concentration is calculated here.

9 Directional transport interpretation

Low forward permeability and directional asymmetry present different follow-up questions from uniformly weak transport. The reported efflux ratio describes this cell system; it does not identify a specific transporter.

No inhibitor-rescue or transporter-isolation result is supplied. The permeability-limited branch designation therefore remains an operational triage conclusion. It should not be expanded into a mechanistic assertion.

10 Kinetic availability interpretation

DMSO dilution can produce transient supersaturation or precipitation. The kinetic result tests availability after dilution and should not be treated as a thermodynamic constant.

Low transport deserves particular caution when dissolved availability is also limited. Conversely, adequate kinetic solubility does not guarantee entry through a monolayer. Reading the tables together supports separation of those constraints.

11 Bioanalytical responsibility

Javier Quintana is the mass-spectrometry lead. The analyte card and assay worksheet retain parent assignment and matrix-specific analytical details. This report releases derived property tables rather than reproducing individual acquisition traces.

Plasma, buffer and transport compartments can differ in analytical response. Sharing a compound identifier does not make their measurements interchangeable. Matrix-specific interpretation is part of the basis for assessing recovery.

12 Analytical limits

Numerical quantitation limits, calibration ranges and individual sample concentrations are not present in the supplied final record. They are not inferred here. A small unbound percentage does not establish unlimited sensitivity in the buffer measurement.

The same limit applies to low receiving-compartment signal and censored solubility values. Replacing a limit with an exact result would change the evidence at the exposure gate. The reported precision is retained without additional digits.

13 Matrix acceptance and exceptions

Helen Marchwood is responsible for bioanalytical quality. The companion assay worksheet records matrix acceptance checks. Recovery columns remain visible in the released binding and transport tables.

No named sample-level acceptance exception, numerical recovery threshold or repeat-run history is supplied in the final tables. None is reconstructed in this report. This qualification does not imply an unreported failure; it defines the scope of the released narrative.

14 Integrated branch assessment

Interpretation begins with dissolved availability, then considers directional transport, and retains human plasma binding as a distinct constraint. This order avoids using favorable binding to mask poor availability or weak flux.

Kelbravik separated a permeability-limited branch from a solubility-limited branch and commissioned a focused formulation assessment. That separation is the central outcome. It does not establish that the same intervention will benefit both branches.

15 Formulation assessment remit

The commissioned work should address availability in the solubility-limited branch. It should distinguish a change in dissolved material from a change in analytical recovery and preserve the compound and accession-lot links.

This report does not specify a vehicle composition or report a formulation improvement. The later assessment needs its own record. Its interpretation should address the constraint identified here rather than an assumed gain in every exposure determinant.

16 Permeability follow-up remit

The permeability-limited branch presents a cellular transport question. Follow-up should retain both directions and recovery, so that changes in availability are not mistaken for changes in barrier transport.

The present work does not establish transporter identity or clinical absorption. Its contribution is to separate plausible constraints within the submitted set, allowing subsequent work to address the appropriate branch.

17 Binding context for later exposure

Human $f_{u,p}$ should accompany the other property tables into the DMPK memo. It will be relevant when later total plasma exposure is interpreted, but no exposure measurement is included here.

Binding does not convert nominal assay concentration into a measured in vivo free concentration. Preserve the percentage basis and recovery record whenever the values are transcribed for a later calculation.

18 Final client decision

Kelbravik separated a permeability-limited branch from a solubility-limited branch and commissioned a focused formulation assessment. Megan Larkin records this as the final outcome of MV-2008-013 for Kelbravik Therapeutics ApS.

The conclusion is an in vitro triage decision. No oral bioavailability, dose proportionality or circulating exposure result is asserted. The matrix and analytical-limit qualifications remain part of the decision basis.

19 Record control and assumptions

The companion record is MV-2008-013_AssayWorksheet_032108_v03.xls. This final R1 interpretation supersedes the 03/05/2008 interim communication for closeout. The original project and accession identifiers remain the links to the underlying assay record.

- Join systems by complete compound identifier.
- Preserve solubility limit symbols.
- Keep recovery beside binding and directional transport.
- Do not assign a biological species to buffer solubility.
- Do not infer an in vivo exposure result from the property set.

Name	Role	Signature	Date
Megan Larkin	DMPK Study Director	<u>M Larkin</u>	03/21/2008
Javier Quintana	Mass-Spectrometry Manager	<u>J Quintana</u>	03/21/2008
Helen Marchwood	Bioanalytical Quality Manager	<u>H Marchwood</u>	03/21/2008